P1: Process Book

Davis Germain

Touch-Sensitive 3D Surface

Technical Advance

Objects that can be coated in conductive paint and instrumented to measure changes in conductivity can sense when and where they get touched by a person's skin.

Airbnb Host and Guest Data

Dataset

Airbnb collects information on guests and hosts across the world. This can be used to make inferences about the users or the platform as a whole.

Smartphone LiDAR

Platform

Apple and other smartphone makers have begun to add LiDAR to phones. LiDAR (Light Detection and Ranging) uses a LASER to emit light and sense its reflection to "see" 3D objects in a scene.

Touch-Sensitive 3D Surface

I enjoyed ideating for this idea the most because it seemed just outside of what was already a real technology. While I did have a fundamental misunderstanding of the idea as a whole, which led to most of my ideas being nullified, it was still good practice for the next 2 ideas. Since I had the wrong idea about the project prompt, all of my ideas are about a product that could scan a large 3D surface and port it to some virtual reality where the user could interact and touch everything as it if was real. I didn't use the AI tools for any of my ideating, it was also difficult coming up with so many ideas. Overall, however, this was the easiest to ideate because I thought of many different applications and markets that could find a way to use a touch-sensitive 3D surface.

As shown in the coming slides, I chose to rank my slides based on cost, difficulty to implement, and overall market receptiveness to the product. The lower the number, the "better" a product was (i.e. cheaper, easier to implement, or more likely to be accepted).

Touch-Sensitive 3D Surface

Matchmaking for Touch-Sensitive 3D Surface

Ranking (1 to 5)

	Specific Capability	Domain and/or Activity	Customer	Application	Financial viability (Cheap to Expensive)		Technical feasibility (Easy to Difficult to implement)		Desire/Acceptance (Immediately Accepted to Won't be Accepted)		Rank
1	Used to explore a 3D space or simulation interactively	Medical	Hospitals	Used to map someones organs to an interactive 3D space to see what's going wrong in more depth	2	All you need is to scan someone's body, which we can already do	1	We can already take various scans of people's bodies	1	This would definitely save lives	4
2		Education	Math Teachers	Can be used to help students visually interact with graphs and shapes to learn geometry	4	Most schools don't get that much funding, and teachers get even less to pay for this product/service	4	You need to be able to dynamically graph shapes and graphs in 3D, which is already difficult without making it interactable	1	Students would use this to learn and it would help visual learners	9
3			Historians	Relive significant moments in history and walk around to explore them.	4	Most schools don't get that much funding, and teachers get even less to pay for this product/service	2	We already have 3D modelling capabilities, you just need to make them interactive	3	There may be some controversy about reliving disasters or attacks because it could affect the student's mental state	9
4		Manufacturing	Manufacturing Companies	Visualize factory layout to find optimizations.	5	It would cost a lot of money to create a device to scan an entire factory	5	Depending on how complex the factory is, this could have to be a very large model	2	Companies are always looking for ways to save money	12
5		Engineering	Tools companies	"Blow up" tools that you can activate or pause to see how they work on the inside/diagnose issue (NOTE: New idea: handicap accesible tools that trigger on a tap or hover)	2	They only need to scan a small tool and are already using standardized parts	1	You can use the battery as part of the tool already	1	Accesibility with tools and crafting would be appreciated	4
6		Real Estate	People looking for houses	Download and interact with it in 3D before buying it.	4	You'd need to be able to scan whole rooms which would need more funding for tools that home buyers may not have	5	Scanning a whole house and mapping it to a big enough space to interact with would be demanding	3	Homeowners may not offer their 3D models up to customers if they are concerned about the state of the house	12
7		Military	Weapons Developers	Weapons testing and seeing the impact of weapons without having to actually use them.	5	You'd need servers or somewhere to run the weapons testing which would need cooling, power, and the capability to do so	5	Implementing physics, chemistry, biology, and every other system the world works with would be immensely difficult	5	Large ethical issue with the public	15
8		Health and Beauty	Make up companies	Allow the user to test makeup on themselves and put it on to see if they like it.	3	Make up companies have the money and incentive to invest in this	2	You'd just need to scan someone's head which already somewhat exists	1	Companies would sell much more makeup because people could get exactly what they want and it would enable iffy shoppers to purchase exactly what they want	6
9		Business	Ad teams	Let viewers try item or see it in physical use before buying.	3	Variable depending on the product	4	You'd need to make the real product and the virtual product before releasing	5	See technical feasability. Companies would not do this	12

Touch-Sensitive 3D Surface

10		Gaming	Game development studios	A new way to make VR games that the user can interact with more naturally		1	VR technology already exists, they just need to add to it	3	This may be difficult to create while overcoming motion sickness	1	Any VR player would purcahse these games, so studios would be inclined to make them	5
11		Agriculture	Farmers	Used as a learning tool to see through the ground and understand how plants grow and when to harvest for best results.		1	Crops are small and you wouldn't need to implement much	2	Crops are small, and the sample size you're growing the crop in won't be large	4	Farmers are probably farmers because they are passionate and know about crops, they wouldn't need a simulation to tell them how	7
12		Policing	Police Departments	Used to review crime scenes from months or years ago for unclosed cases without having to section off the scene in the real world for that long.		2	Government can pay for this	3	You'd need to scan potentially whole rooms or buildings with complex piping systems	5	Very controversial with the public	10
13		Traffic Control	Civil Engineers	Interact with and "live" in their changes to traffic flow before shutting down lanes for years to make sure they are actually useful.		2	It wouldn't cost too much to simulate cars since there are already racing games and practice for competitve racing	2	There's no need for physics or anything outside of the car itself, so the rest can be a naive or incomplete solution	4	Doesn't seem entirely necessary, civil engineers can just observe and run simulations without having to live in them	8
14		Weather	Weather Companies	See the damages a natural disaster will cause before it happens to evacuate the people affected the most.		4	It would be very expensive to simulate and run all that damage in an interactive environment	5	It would be hard to run all the calculations to predict debris and the total damages	3	With decreased funding, it would be hard to predict a natural disaster that far in advance, let alone exactly the path it will take	12
15		Accesibility	People with disabilities	If they cannot perform more complex tasks or interactions, they can use simple taps or interactions combined to make a more complex task representable		1	It would be easy to map a combination of interactions to a more complex interaction	2	The biggest difficulty is thinking of combinations to map to more complex interactions and communicating that to the user	1	Accesibility is hardly a bad thing	4

Touch-Sensitive 3D Surface

After getting critique from my classmates, this was when I realized that what I had ideated for and prompt itself were two different things. I took inspiration from my ideas about the prompt I used to create a couple new ideas for the class prompt. The critique I received also opened my eyes to whether a product was marketable at all. I had been so focused on whether something was feasible that I forgot to acknowledge whether a product would actually sell or not, and who would it affect. This led me to select my final idea: handicap accessible power tools.

Handicap accessible power tools seemed like the obvious best pick because everyone likes having their own tools and feeling confident in their own abilities, however people with disabilities are barred off from this feeling because of something they couldn't control. By coating the power tools with the touch-sensitive paint, it clears the way for a variety of applications allowing people with disabilities to use power tools. An example of this is coating drill triggers in the paint. People who may have lost motor function in their hand or may not be able to grip and pull wouldn't need to anymore. They could rest their finger in the trigger's grip and lightly rest their finger on the trigger to start the drill.

Airbnb Host and Guest Data

Because I did not use any of the AI tools for ideating, thinking of applications and inferences from the data was very hard. It was hard to imagine who would want to use Airbnb housing data outside of the guests, homeowners, and Airbnb themselves. Despite this, I still came up with a few ideas that I was happy with, in markets ranging from Education and Energy to Construction and Entertainment. I enjoyed ideating for this dataset in a unique way, because I wasn't given the product or the core technology as part of the prompt. We were given the data that the product would rely on and had to come up with ideas using the data. This new way of ideating was interesting in it's own way, albeit difficult.

Similarly to the Touch-Sensitive 3D Surface, I ranked my ideas based on cost to implement, difficulty to implement, and market receptiveness.

Airbnb Host and Guest Data

Matchmaking for Airbnb Host and Guest Data

	Specific Capability	Domain and/or Activity	Customer	Application	Financial viability (Cheap to Expensive)		Technical feasibility (Easy to Difficult to implement)		Desire/Acceptance (Immediately Accepted to Won't be Accepted)		Rank
1	Predicting where regions of people go for breaks	Travel	Airlines	Knowing whether they can raise or lower plane tickets depending on demand	1	Dynamic pricing already exists	2	The data is already provided, and AI could easily map it across a globe to make these predictions.	3	Dynamic Pricing is generally a little controversial because people don't like being squeezed for every penny.	6
2		Housing	Homeowners	Knowing whether they should post a house for rest or not over a given period	1	Many companies already have dynamic pricing	2	The data is already provided, and AI could easily map it across a globe to make these predictions.	3	Dynamic Pricing is generally a little controversial because people don't like being squeezed for every penny.	6
3		Construction	Construction companies	Knowing when a region will be most empty to perform construction at that time	3	It may cost the company to delay construction efforts	2	The data is already provided, and AI could easily map it across a globe to make these predictions.	4	People with less money or who can't travel would be heavily punished for reasons that may be out of their control	9
4		Business	Small businesses	Knowing if they will or won't have business to keep the shop running	2	It would save the company money, but they may miss out on big customers	2	The data is already provided, and AI could easily map it across a globe to make these predictions.	3	Dynamic Pricing is generally a little controversial because people don't like being squeezed for every penny.	7
5		Energy	Energy providers	Knowing when rolling blackouts will have the lowest impact on an area	3	Rolling blackouts are usually associated with an issue to diagnose, so they may not want to wait	2	The data is already provided, and AI could easily map it across a globe to make these predictions.	4	People with less money or who can't travel would be heavily punished for reasons that may be out of their control	9
6		Policing	Police departments	Knowing when certain areas are less inhabited to patrol less often	1	It would be free to reallocate where police roam	2	The data is already provided, and AI could easily map it across a globe to make these predictions.	5	If people know places are policed less often, they'll just rob there instead	8
7		Entertainment	Weather dependant activity companies	Knowing how many people are coming for a season to prepare in advance	2	It wouldn't be hard to develop AI to make these predictions, and on top of that getting a good estimate of the customer count before it happened would help them not under/over buy tools for their product	2	The data is already provided, and AI could easily map it across a globe to make these predictions.	1	People would appreciate companies being prepared for them	5
8	Determining the attractiveness of a rental	Housing	Homeowners	Homeowners can know what they're doing right or wrong with their rental to improve sales	1	This data wouldn't be hard to extract or pay for, and the biggest cost would be buying the new furniture or things to make their house more appealing.	1	This can easily be assumed from data showing sales and comparing those sales between various rentals in a given area	1	Homeowners could improve their house to get more sales, home renters could expect an increase in quality of what they are renting	3
9		Construction	House construction	Construction companies and plot owners know exactly what sells in that area to make the most desireable house	2	It wouldn't be hard to implement but sales may go down slightly so companies wouldn't restrict themselves to just building ideal renting houses	1	This can easily be assumed from data showing sales and comparing those sales between various rentals in a given area	4	If you're always constructing the "perfect" house, some people may complain that you aren't making anything unique and the whole point of rental apps would disappear if competition was always equal	7

Ranking (1 to 5)

Airbnb Host and Guest Data

10		Education	Psychologists	They can use this data to understand what people like and why they like it to learn more about human thinking.	1	Most psychologists in this area would accept this data and it wouldn't cost anything aside from psychologist manpower	1	This can easily be assumed from data showing sales and comparing those sales between various rentals in a given area	2	If psychologists get consent, not many people complain with participating as long as it doesn't harm them	4
11	Predicting total income during a period	Housing	Homeowners	People can use this to know how much they should expect to make	1	Homeowners would love this data so they can plan their own expenses	2	Given the data of previous years/seasons and each rental "type", it wouldn't be hard to make these assumptions given no extraneous events	1	This kind of data probably wouldn't get back to customer and would just be used for the renter	4
12			Home renters	People can use this to know how much they should expect to pay	1	Home renters would love this data so they can plan their own expenses	2	Given the data of previous years/seasons and each rental "type", it wouldn't be hard to make these assumptions given no extraneous events	2	This data would also stay within the customer, but it might somehow affect renters and their profits which would make them either lower prices to be more competitive or settle for less money	5
13		Business	Rental platforms	They can use this information to know how much to expect people to rent	1	Business would like this to know what to expect for profit and also learn how to push their algorithm to maximize this	2	Given the data of previous years/seasons and each rental "type", it wouldn't be hard to make these assumptions given no extraneous events	5	People don't enjoy being "penny pinched", so if any significant changes to the algorithm were made that forced people to either spend money or get off the app then they would be upset.	8
14	Predicting buyers intentions	Housing	Home owners	They can predict if someone is just entertaining an idea or actually intends to rent	2	This wouldn't be too expensive, although it would lower sales which would negatively impact the renter and platform	4	This would be harder to predict because you would also need to know how long a given person usually views a place for before buying, on top of tracking when the app is closed or if they leave it on on accident	4	Some people just like to "browse" or may accidentally leave their phone open. These people would be pushed away from the app, making the experience worse for them.	10
15		Business	Rental platforms	They can change their algorithm to stop recommending places that customers just look at and replace it with real purchases to promote spending	1	Business would like this to know how to push their algorithm to maximize profits	4	This would be harder to predict because you would also need to know how long a given person usually views a place for before buying, on top of tracking when the app is closed or if they leave it on on accident	5	People don't enjoy being "penny pinched", so if any significant changes to the algorithm were made that forced people to either spend money or get off the app then they would be upset.	10

Airbnb Host and Guest Data

The biggest thing that critique from classmates opened me up to was the market's "need" for this product. No matter how interesting or applicable the idea was, if the market didn't want it then it would flop. It also reminded me that a lot of companies are driven by the profit and not the ethics or morals. For example, many of my ideas would cost the company money and time which I was fine spending because I was just ideating and I didn't work for the company. However if I was working at the company, these precious resources that I was spending so easily wouldn't have been as easy to spend.

While my selected idea did not come from this category, I still had two ideas I liked from this category that I brought up for critique. The ideas I brought up weren't the highest ranking according to the metrics I explained earlier, they were novel ideas that I was proud of and would've enjoyed pursuing. Using the housing data for things like timing construction and rolling blackouts is a good idea that would raise overall customer satisfaction with the company. I didn't end up going with these ideas in the end because it would've cost the company lots of money which was out of the question while also creating an unfair advantage to families who could travel over those who couldn't.

Smartphone LiDAR

As with the other ideas, I did not use any AI tools for ideating with the smartphone LiDAR. This was, by far, the most difficult prompt to think of ideas for. If I had done one thing differently with this project, it would have been using the Vercel tool to create ideas for this prompt. Because smartphones having LiDAR is common, it was hard to break through the barriers of reality to think of new or innovative ideas using LiDAR. Going back and using the AI tool would have helped me create better ideas while also preserving the fun and teaching lesson received from this whole project.

Because most of my ideas were already either real or easily achievable, the rankings I used in with this prompt functioned a little differently. Even though the categories were still the same, most ideas earned either a one or two on financial and technical feasibility. Because of this, the most influential category was the market receptiveness. Given that all else was equal, this metric dictated where, on the limited scale, each idea ranked.

Smartphone LiDAR

Matchmaking for Smartphone with LiDAR

Ranking (1 to 5)

	Specific Capability	Domain and/or Activity	Customer	Application	Financial viability (Cheap to Expensive)		Technical feasibility (Easy to Difficult to implement)		Desire/Acceptance (Immediately Accepted to Won't be Accepted)		Rank
1	Taking better photos	Business	Phone companies	They can market these new technologies to increase sales	1	This is cheap to do with $26 as proposed	1	This is just the base technology which already exists	2	No one would be hurt by this except maybe allowing phone companies to mark phones up as more expensive than they may be worth	4
2			App developers	They can use these technologies to improve gameplay	1	This is cheap to do with $26 as proposed	1	This feature is already built into most phones for developers to use	1	Nobody would be hurt by this, and companies would gain player bases for new games with this technology	3
3		Social Media	Social media apps	The new technology will create better photos that people will want to post	1	This is cheap to do with $26 as proposed	2	This feature is already built into most phones for developers to use, the apps would just have to make sure to implement this and read the data LiDar spits out	2	This technology would make these apps money, but there would be slight push back because not everyone loves social media apps and it would be more demanding on users phones to load these images	5
4			Social media users	The new technology will create better photos that people will want to post	1	This is cheap to do with $26 as proposed	1	This feature is already built into most phones and on by default for users	1	People want to post the best picture they have, so it would not hurt anyone to be posting these picture	3
5		Policing	Crime Scene investigators	They can use this improved technology to scan and take better more detailed pictures of a crime scene	1	This is cheap to do with $26 as proposed	1	This is just the base technology which already exists	4	This would be a little interesting, because if everything was stored on one phone for pictures then it might be easy to breach or hack and remove those photos	6
6	Scanning for 3D models	Game and Media	Game development studios	They can use this technology to "bring games to home" and let kids use household items they know and love to play games with	2	This is slightly more difficult because you have to scan around a whole object, but also very cheap as a whole with LiDar	2	The data would be provided by the user, and it would be easy for AI to put this into the game	3	Parents would be concerned about increased game time which may lower player base, as well as the concern about collecting the data these kids given about their house to sell to ad companies	7
7		Education	Architecture	They can use this technology to find an easy way to submit their assignments if they aren't in person or need to stay home	2	This is slightly more difficult because you have to scan around a whole object, but also very cheap as a whole with LiDar	1	This would be an easy 3D scan which is already implemented in many cameras or apps	1	This would allow students to relax and keep themselves safe when sick knowing they can submit accurate representations of their work from home	4
8	More accurate measurements	Construction	Construction companies	Construction companies can use this to reduce the number of tools they have to buy or carry around when making smaller objects or rooms	1	This is easier than scanning for models, and it would not change the user base because they already have the tools in case they don't want this solution	1	The scanning would be easy to perform because its built in to many phones	5	Many construction workers were trained and like using these tools, so I think if this type of idea was implemented, it wouldn't gain much traction and people wouldn't use it	7

Smartphone LiDAR

9		Housing	Home owners	Home owners would use this to see how much space they have and how big different furniture items are	1	This would allow home owners to spend less money to achieve the same goal	2	The scanning already exists, a company would just need to be able to take this ability and put it into an app that can adapt and suggest furniture to different audiences	3	People may be concerned about collecting data on household furniture and layout, but for the most part this would just serve to increase user base	6
10		Education	Teachers	Math or science teachers would have their students use this to get more precise measurements and getting good lab results	2	Schools usually already have these tools, however this would require phones in school which is a controversial point	3	This type of information would need to interact with liquids or moving objects potentially, so it would require a slightly more advanced technology that may be hard if phones aren't available in class	5	Having phones in school is a controversial topic that would require some serious discussion so its not a distraction	10
11	New AR technology	Games and Media	Game development studios	This kind of technology led to apps like Pokemon GO, so companies would always use this	2	This falls under a similar category as performing 3D models, which we can already do for cheap and if users don't like a game it doesn't hurt them to not play it	2	This technology already exists within apps like Pokemon GO, so companies would only need to create their own version and then can reuse it across apps	1	This would raise more money for game development companies to explore this space, leading to more games using LiDar that people can shop for	5
12			Phone users	People with phones would appreciate this because it monetizes or incentivizes them to go outside and exercise more	2	This would already be a part of most phones and would incentivize people to go outside more, so they'd be slightly more inclined to get this phone	1	They don't need to implement anything, they just need the app development companies to release the apps	1	In 2016 there was a large push for more apps like this to increase activity outside, so clearly users like these	4
13	Home Decoration	Manufacturing/Construction	Furniture companies	Furniture companies can use this to help sell their goods to people if they can picture it in an ideal home	1	This would be fairly cheap to implement and would increase profits as long as they had good products that people wanted to buy	3	You would need a way to store the information about a piece of furniture and also be able to "place" it in a virtual environment for the user to look at	1	This would increase purchaser confidence and also allow furniture companies to sell more, giving them more reign for creativity with the knowledge that someone will buy their product with confidence and not return it	5
14		Housing	Home owners	Home owners can use this to see if decorations look good or achieve the goal they want	1	This technology is already in most phones, so it would be cheap for homeowners to use to make better purchases	1	This technology already exists on phones, but the user is reliant on app development studios to create apps that they can use compatible with the furtniture companies	1	Furniture companies already do this so they gain sales from this, and it allows users to be more confident in their purchases	3
15	Letting people "live" in or experience a room or house	Housing	House renters or buyers	By being able to live in it beforehand, it can make them more certain about their purchases	3	This would be harder because you have to scan a much larger area, but it would be cheap to implement and just take more time to perform	5	It would be very intense to scan an entire house and keep track of it correctly while filling in any blanks that may have not been scanned or missed	4	It would be more work for the house renter and also house buyer to set up properly, so people would just prefer to tour houses in person so they can also ask questions to the landlord at the same time	12

Smartphone LiDAR

The two biggest pieces of critique I received were about the customer base and uniqueness of my ideas. My first idea, which was about using the new technology to create more detailed pictures so architecture students could submit pictures of their work online and still be graded fairly, was very specific. The number of architecture students is already very low, and even lower is the number of them who regularly get sick around submission dates for their assignments. Most of the time professors could either extend the deadline or work something else out with the student, so the first idea would be hard to actually realize. The other idea would let people visualize or “place” furniture in their house through AR on their phones to see if it fit or looked good. Since this already exists, it didn’t feel innovative or creative to use this idea as my main selected concept, so I decided to leave it behind for other ideas.

I chose these two ideas because both of them felt either new or useful respectively. This didn’t turn out to be the most helpful metric, however, because one of them was too unique and the other was too useful (so it was already implemented). Both ideas also scored low numbers according to the metrics I used, so I knew the ideas would be well received in their respective customer bases and be feasible to implement.

Pitch Slide

HandiTools: Handicap Accessible Power Tools

Davis Germain

Capabilities

- Touch-sensitive surface to detect when an area (i.e. a trigger) of a power tool has been pressed.
- Allows people with physical disabilities to use power tools

Co-Creation of Value

People with disabilities: Can use power tools to create DIY or home projects

Businesses: Increases customer base and company reputation

“Drunk Island”

This product is as easy as replacing the trigger or button with spray paint. On drunk island, people could easily take off the button and spray the paint where the piece used to be because both are simple instructions.

Feasibility: Rank #1

Financial: 1/5

¼ Americans have disabilities (CDC), and the spray paint can be applied for cheap (CMU)

Technical: 1/5

Touch sensing, which is a basic capability of the root product, and rerouting of power (Drill Ex.)

Acceptance: 1/5

Customers with physical disabilities would purchase this to work on other projects. Other customers would continue to buy standard products.

Personal Reflection

Overall, I think I benefited from this project greatly. Matchmaking is a completely new way to ideate or find solutions and uses to technology that I hadn't used before, and I really enjoyed testing it out with this project. While I did struggle to self-generate 45 unique ideas, I am still happy with the number of ideas I came up with and I think most of them are creative and applicable to fields I didn't think any of this technology could be applied to. I will be sure to use matchmaking as a innovation approach in future projects.

I did not use the AI Matchmaking tool, but from what I saw when I tried it in class, it did not seem very helpful long-term. By this I mean that it probably would've been helpful for me to use during this project to come up with new ideas, but after someone follows the matchmaking process a handful of times, I would think they have their own way to thinking of new applications for a technology that would be more accurate than what the AI gave back. For example, I am not proud of a couple of my ideas regarding the LiDar, so AI may have been helpful to generate new ones with that technology. However, if I used matchmaking for 2 or 3 other projects and then came back to the LiDar, I think that I would be able to create 15 new ideas that would be better than what the AI tool would give me.

Thank You!

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